DATA DEFINITION LANGUAGE (DDL) :

DDL IS USED TO CREATE / BUILD / CONSTRUCT AN OBJECT IN THE DATABASE & IT ALSO DEALS WITH THE STRUCTURE OF THE TABLE.

TYPES OF DDL STATEMENTS

1. CREATE
2. RENAME
3. ALTER
4. TRUNCATE
5. DROP

CREATE

IT IS USED TO CREATE STRUCTURE OF A TABLE / OBJECT.

SYNTAX:

```
CREATE TABLE TABLE_NAME
(
COL_1 DATATYPE  CONSTRAINTS,
COL_2 DATATYPE  CONSTRAINTS,
COL_3 DATATYPE  CONSTRAINTS,
  :             :                 :
  :             :                 :          ,
COL_n DATATYPE  CONSTRAINTS
);
```

DESC TABLE_NAME;

EX:

```
CREATE TABLE DEMO
(
ID VARCHAR(4)  PRIMARY KEY,
NAME VARCHAR(20) NOTNULL ,
SAL NUMBER(8,2) NOT NULL CHECK(SAL)>0
);
```

NOTE :

> IF WE WANT TO ADD ONE MORE COLUMN GIVE COMMA (' , ')

> IN ORDER TO PASS MULTIPLE CONSTRAINTS JUST GIVE SPACE

RENAME

IT IS USED TO CHANGE EXISTING TABLE_NAME

SYNTAX:

```
RENAME EXISTING_TABLE_NAME TO NEW_TABLE_NAME;
```

EX:

```
RENAME DEMO TO DEMO7
```

ALTER

IT IS USED TO MODIFY THE STRUCTURE OF THE TABLE

SUCH AS

- TO ADD A COLUMN
- TO REMOVE A COLUMN
- TO RENAME A COLUMN
- TO MODIFY THE DATATYPE
- TO MODIFY THE CONSTRAINTS
- TO ASSIGN A FOREIGN KEY

- **TO ADD A COLUMN**

SYNTAX:

```
ALTER TABLE TABLE_NAME
ADD COL_NAME DATATYPE CONSTRAINTS;
```

EX:

```
ALTER TABLE DEMO7
ADD PH_NO NUMBER(10) UNIQUE NOT NULL CHECK(LENGTH(PH_NO)=10) CHECK(PH_NO>0);
```

- **TO REMOVE A COLUMN**

SYNTAX:

```
ALTER TABLE TABLE_NAME
DROP COLUMN COL_NAME;
```

EX:

```
ALTER TABLE DEMO7
DROP COLUMN PH_NO;
```

- **TO RENAME A COLUMN**

SYNTAX:

```
ALTER TABLE TABLE_NAME
RENAME COLUMN EXISTING_COL_NAME TO NEW_COL_NAME;
```

EX:

```
ALTER TABLE DEMO7
RENAME COLUMN SAL TO SAL_7;
```

› TO MODIFY DATATYPE

SYNTAX:

```
ALTER TABLE TABLE_NAME
MODIFY COL_NAME NEW_DATATYPE;
```

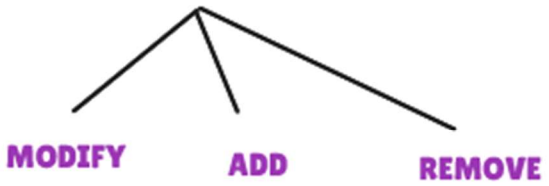
EX:

```
ALTER TABLE DEMO7
MODIFY NAME CHAR(25);

ALTER TABLE DEMO7
MODIFY NAME CHAR(30);

ALTER TABLE DEMO7
MODIFY NAME VARCHAR(30);
```

› TO MODIFY CONSTRAINTS



SYNTAX:

MODIFY BOTH DATATYPE AND CONSTRAINT

```
ALTER TABLE TABLE_NAME
MODIFY COL_NAME EXISTING_DATATYPE NEW_CONSTRAINT;
```

EX:

```
ALTER TABLE DEMO7
MODIFY NAME VARCHAR(30) NULL;

ALTER TABLE DEMO7
MODIFY NAME VARCHAR(32) NOT NULL;
```

› TO ADD CONSTRAINT

SYNTAX:

```
ALTER TABLE TABLE_NAME
ADD CONSTRAINT CON_REF_NAME CONSTRAINT_TYPE(COL_NAME);
```

CON_REF_NAME: It can be any name But it MUST BE UNIQUE

It is the combination of Constraint_Type and Col_Name

EX: EMAIL ADDING UNIQUE -----> UNI_EMAIL

EX:

```
ALTER TABLE DEMO7
ADD EMAIL VARCHAR(40) NOT NULL;

ALTER TABLE DEMO7
ADD CONSTRAINT UNI_EMAIL UNIQUE(EMAIL);
```

› **TO REMOVE CONSTRAINT**

SYNTAX:

```
ALTER TABLE TABLE_NAME
DROP CONSTRAINT CON_REF_NAME;
```

EX:

```
ALTER TABLE DEMO7
DROP CONSTRAINT UNI_EMAIL ;
```

HOW TO GET CON_REF_NAME

```
SELECT *
FROM USER_CONSTRAINTS
WHERE TABLE_NAME = 'T_Name';
```

TO ASSIGN FOREIGN KEY

STEP 1:
```
ALTER TABLE TABLE_NAME
ADD COL_NAME DATATYPE;
```

STEP 2:
```
ALTER TABLE TABLE_NAME
ADD CONSTRAINT CON_REF_NAME FOREIGN KEY(COL_NAME)
REFERENCES PARENT_TABLE_NAME(COL_NAME);
```

EX:
```
ALTER TABLE DEMO7
ADD DEPTNO NUMBER(2);

ALTER TABLE DEMO7
ADD CONSTRAINT FK_DEPTNO FOREIGN KEY(DEPTNO)
REFERENCES DEPT(DEPTNO);
```

WHILE CREATING TABLE WE CAN ADD FOREIGN KEY DIRECTLY

STEP 1: CREATE PARENT_TABLE TEACHER(TID, TNAME)

STEP 2: CREATE CHILD_TABLE STUDENT(SID, SNAME)

STEP 3: Now directly we are adding Relationship between table,
But we need to ADD A COLUMN WITH DATATYPE IN CHILD DATATYPE

STEP 4: ADD CONSTRAINT

CREATION OF PARENT TABLE

```
CREATE TABLE TEACH7
(
 TID VARCHAR(2) PRIMARY KEY,
 TNAME VARCHAR(20) NOT NULL
);
```

WHILE CREATING A CHILD TABLE ESTABLISHING RELATION

```
CREATE TABLE STUD7
 (
 SID NUMBER(2) PRIMARY KEY,
 SNAME VARCHAR(25) NOT NULL,
 TID VARCHAR(2),
 CONSTRAINT FK_TID FOREIGN KEY(TID) REFERENCES TEACH7(TID)
 );
```

› TRUNCATE

IT IS USED TO REMOVE ALL THE RECORDS / DATA FROM THE TABLE PERMANENTLY

SYNTAX:

```
TRUNCATE TABLE TABLE_NAME;
```

EX:

```
TRUNCATE TABLE DEMO7;
```

› DROP

IT IS USED TO REMOVE THE TABLE FROM THE DATABASE

SYNTAX:

```
DROP TABLE TABLE_NAME;
```

RECYCLE BIN

EX:

```
DROP TABLE DEMO7;
```

FLASHBACK :

IT IS USED TO RESTORE THE TABLE FROM THE RECYCLEBIN

SYNTAX:

```
FLASHBACK TABLE TABLE_NAME
TO BEFORE DROP;
```

EX:

```
FLASHBACK TABLE DEMO7
TO BEFORE DROP;
```

PURGE:

IT IS USED TO REMOVE THE TABLE FROM THE RECYCLEBIN

SYNTAX:

```
PURGE TABLE TABLE_NAME;
```

EX:

```
PURGE TABLE DEMO7;
```

```
SHOW RECYCLEBIN;
```